
Understanding the Risks of Repeated Corticosteroid (Steroid) Injections for Pain

What Are Corticosteroid Injections?

Corticosteroid injections (commonly called "steroid injections" or "cortisone shots") are a widely used treatment for pain and inflammation in joints, the spine, and soft tissues. They work by reducing inflammation at the site of injection and can provide significant short-term pain relief.

Common corticosteroids used in pain procedures include:

- **Depo-Medrol** (methylprednisolone acetate)
- **Kenalog** (triamcinolone acetonide)
- **Dexamethasone** (Decadron)

While these injections can be very effective, repeated use carries important risks that you should understand.

Risks of Repeated Corticosteroid Injections

Effects on Your Bones (Osteoporosis and Fractures)

Repeated steroid injections can reduce bone mineral density (bone strength), increasing the risk of fractures. This risk is especially important for:

- Postmenopausal women.
- Older adults.
- People already diagnosed with osteoporosis or osteopenia.

Studies show that postmenopausal women who received multiple injections (with cumulative triamcinolone doses of approximately 400 mg) had lower bone density in the hip. Even a single epidural steroid injection can temporarily decrease bone density for 6 to 12 months.

Adrenal Gland Suppression

Your adrenal glands produce cortisol, a hormone essential for responding to stress and illness. Steroid injections can temporarily suppress your body's natural cortisol production. Research has shown:

- Adrenal suppression can occur within 2 days of a single injection.

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- The hypothalamic-pituitary-adrenal (HPA) axis may be suppressed for 1 to 4 weeks after a single intra-articular injection.
 - Higher steroid doses cause greater suppression.
 - It is recommended to allow a 2-to-4-week gap between any two steroid injections to allow the adrenal glands to recover.

Blood Sugar Elevation

Steroid injections can raise blood sugar levels, even from local injections. This is particularly important if you have diabetes. Blood sugar levels may rise within a few days of the injection and can persist for several days. If you have diabetes, speak with your doctor about monitoring your blood sugar more closely after a steroid injection.

Cartilage Damage and Joint Deterioration

Repeated intra-articular (into the joint) steroid injections have been shown to accelerate cartilage loss. Higher doses and longer duration of treatment increase this risk. Emerging data suggest that repeated hip injections may be associated with a rare but significant risk of rapidly progressive osteoarthritis.

Local Side Effect

- **Skin changes:** thinning, lightening (depigmentation), or bruising at the injection site.
- **Soft tissue atrophy:** fat pad or tissue shrinkage near the injection site.
- **Tendon weakening or rupture:** injections near tendons can weaken them, and repeated injections increase this risk.
- **Post-injection flare:** temporary worsening of pain lasting 1 to 2 days, occurring in about 2 to 4% of patients.
- **Infection:** rare but possible at any injection site.

Other Possible Systemic Effects

With repeated injections, you may also experience:

- Temporary facial flushing.
- Temporary increase in blood pressure.
- Weight gain and fluid retention.
- Mood changes or insomnia.

How Often Can Steroid Injections Be Safely Given?

General recommendations suggest steroid injections into any single joint should be limited to **3 to 4 times per year**. A minimum interval of **3 months** between repeat injections into the same joint is commonly recommended.

When receiving injections at multiple sites or from multiple providers, it is important to keep track of your **total cumulative steroid dose** across all injection sites.

Maximum Recommended Yearly Doses

Per Treatment Session (Single Visit) Limits

Current guidance recommends not exceeding the following total body doses **per treatment session**:

Steroid	Maximum Per Session
Dexamethasone	10 mg
Methylprednisolone (Depo-Medrol)	80 mg
Triamcinolone	80 mg

Cumulative Annual Dose Limits

For **postmenopausal women** (the highest-risk group for bone loss), the literature recommends a cumulative annual limit of approximately **200 mg of methylprednisolone equivalents** from all steroid injections combined. Over 3 years, the cumulative limit is approximately **400 mg of methylprednisolone equivalents**.

For **healthy men**, the 3-year cumulative limit for bone density impact is approximately **3,000 mg (3g) of methylprednisolone equivalents**.

Dose Equivalency Table

The following table helps illustrate how common steroids compare. These are the typical per-site doses used in pain injections.

Steroid	Small Joint / Soft Tissue	Large Joint	Relative Potency
Methylprednisolone (Depo Medrol)	2–10 mg	10–80 mg	5x hydrocortisone
Triamcinolone (Kenalog)	10–20 mg	10–40 mg	5x hydrocortisone

Because methylprednisolone and triamcinolone have equivalent potency, their dose limits are essentially interchangeable (i.e., 40 mg of Depo-Medrol $\approx$ 40 mg of triamcinolone in terms of systemic steroid exposure).

Dexamethasone is approximately 5 times more potent than methylprednisolone, so 10 mg of dexamethasone is roughly equivalent to 50 mg of methylprednisolone.

Who Is at Higher Risk?

You may be at higher risk for complications from repeated steroid injections if you:

- Are a postmenopausal woman.
- Are over 65 years of age.
- Have osteoporosis or osteopenia.
- Have diabetes.
- Are receiving steroid injections from multiple providers or at multiple sites - Have a history of avascular necrosis.
- Are scheduled for joint replacement surgery (a corticosteroid-free interval of at least 3 months before surgery is generally recommended).

What You Can Do

- **Keep a record** of all steroid injections you receive, including the type of steroid, dose, date, and injection site.
- **Inform all your healthcare providers** about recent steroid injections, especially if you see multiple practitioners.
- **Monitor your blood sugar** closely after injections if you have diabetes.
- **Report any symptoms** of adrenal insufficiency to your doctor, such as unusual fatigue, dizziness, nausea, or muscle weakness.
- **Ask about bone health:** Discuss whether bone density testing or preventive treatment for osteoporosis is appropriate for you.
- **Follow your doctor's recommendations** about the timing and frequency of injections.

Important Reminders

- Steroid injections are a valuable tool for managing pain when used appropriately.
- The goal is always to use the **lowest effective dose** at the **longest safe interval**.
- Your pain specialist will weigh the benefits of pain relief against the potential risks for your individual situation.
- Never hesitate to ask your healthcare provider questions about your treatment.



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In an emergency

→ Call 911, or visit your local emergency department

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